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ORIGINAL PAPER

The invasive mud crab enforces a major shift in a rocky littoral invertebrate community of the Baltic Sea

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Abstract In rocky littoral communities, intense herbivory allows for the occurrence of trophic cascades where higher trophic levels influence producer communities. Invasive predators can be especially effective in imposing trophic cascades. The North American mud crab *Rhithropanopeus harrisii* is a recent invader in the Baltic Sea, with an expanding distribution range. Here, we document the effects of mud crab on the native invertebrate community associated with the key foundation species *Fucus vesiculosus*. During the initial 3 years of invasion, mud crab abundance in *F. vesiculosus* increased from 2 % to about 25 % of the algae being inhabited by crabs. Simultaneously, the invertebrate community underwent a major transition: Species richness and diversity dropped as a consequence of decreasing abundance and the loss of certain taxa. The abundance of gastropods decreased by 99 % and that of crustaceans by 75 %, while chironomids completely disappeared. Consequently, the community dominated earlier by herbivorous and periphyton-grazing gastropods and crustaceans shifted to a mussel dominated community with overall low abundances of herbivores. At the same time filamentous epiphytic algae prospered and the growth rate of *F. vesiculosus* decreased. We suggest that this shift in the invertebrate community may have far reaching consequences on ecosystem functioning. These arise through changes in the strength of producer–herbivore interaction, caused by mud crab predation on the dominating grazer taxa. This interaction is a major determinant of ecological function of ecosystems, i.e. productivity and energy flow to higher trophic levels. Therefore, the decrease in herbivory can be expected to have a major structuring role in producer communities of the rocky littoral macroalgal assemblages.

Keywords Mud crab · Baltic Sea · Predation · Herbivory · Ecosystem function

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Introduction

Recent research in community ecology has clearly revealed the importance of higher trophic levels in regulating communities and ecosystem functioning. This regulation is caused by trophic cascades, where the abundance of top-predators resonates down to the abundance and community structure of producers (reviewed by Pace et al. 1999; Shurin et al. 2002; Estes et al. 2011; Sergio et al. 2014). Top-down regulation through the trophic levels is particularly pronounced in marine littoral communities (Shurin et al. 2006; Gruner et al. 2008), where the preconditions for cascades to arise—relatively simple and linear food chains with identifiable trophic levels and intense trophic interactions—are commonly present. In particular the producer–herbivore interaction needs to be strong, and predators need to have the capability to affect the population dynamics of their prey. In marine littoral environments, producers, herbivores and predators form distinct trophic levels, herbivory on both macrophytes and microalgae is typically severe (Hillebrand 2009; Poore et al. 2012), and predation has a potential to strongly affect herbivore population dynamics through consumption and/or fear effects (Reynolds and Sotka 2011; Haavisto and Jormalainen 2014).

Invasive predators that are able to settle and become abundant in the community can be especially effective in imposing trophic cascades (Carlton 1989; Snyder and Evans 2006; Paolucci et al. 2013). This is because native species without a common evolutionary history with the novel predator may lack effective anti-predation adaptations against it. Anti-predation adaptations are likely to be absent especially when there are no functionally-equivalent native predators in the community (Sih et al. 2010; Polo-Cavia and Gomez-Mestre 2014). This is evidenced by terrestrial and aquatic habitats where invasive predators have been found to have twice the impact on their prey than their native predators (Salo et al. 2007; Paolucci et al. 2013). In marine habitats, invasive predatory species typically have negative effects on local species richness, on the diversity of their own trophic levels and on those below, caused by the effects of competition and consumption (Thomsen et al. 2014).

Crabs are among the most successful aquatic invaders. Their invasiveness is facilitated by dispersal-supporting life-history traits and tolerance to a wide range of salinities (Hänfling et al. 2011). Invasive predators that are strong interactors, such as omnivorous crabs with a voracious predatory feeding mode, can compete with native predators thus having a pronounced influence on local communities. Invasive crabs have been found to reduce abundances of native invertebrates (Grosholz et al. 2000; Falk-Petersen et al. 2011; Epifanio 2013) and their effects may extend to trophic structures and ecosystem function such as e.g. productivity and fluxes of energy and matter in the food webs (Hänfling et al. 2011).

The Baltic Sea is a biogeographically young and species-poor ecosystem; it is being colonized by invasive species at an accelerating rate (Leppäkoski et al. 2002; Paavola et al. 2005). One of the recent invaders is the North American estuarine mud crab *Rhithropanopeus harrisii*, which has already invaded a number of European estuaries. The first sightings of the mud crab in the Baltic Sea date back to 1936, though its distribution was then confined to the southern and south-eastern Baltic Sea, and it has recently been spreading along the Finnish coast and the Gulf of Riga (Kotta and Ojaveer 2012, and references therein). It was found for the first time in the Finnish Archipelago Sea in 2009 after which it has spread to over 80 known locations about 30 km west, south and south-east of the site of the first sighting (Fowler et al. 2013). Currently, the Archipelago Sea harbors the northernmost known Baltic Sea population of the species. In the Baltic Sea, the species inhabits a large variety of habitats but shows a preference for vegetated habitats and brown macroalgal stands dominated by *Fucus vesiculosus* (Nurkse et al. 2015; Aarnio et al. 2015).

Although the mud crab is recognized by *Finland's National Strategy on Invasive Alien Species* as a potentially locally harmful species that requires monitoring, so far very little is known about its ecological role or influence on the local communities. As there are no native brachyuran crabs in the Northern Baltic Sea, this functionally novel predator can potentially be expected to have a strong effect on benthic communities, particularly on its preferred prey species. The mud crab is known to feed on several common littoral mollusks and crustaceans (Hegele-Drywa and Normant 2009; Forsström et al. 2015). It has a flexible and omnivorous feeding pattern, the composition and breadth of its diet varying with the availability of prey (Hegele-Drywa and Normant 2009) as well as with the life-history stage (Aarnio et al. 2015).

Here we document, for the first time, the effect of mud crabs on the rocky littoral invertebrate community associated with bladderwrack *F. vesiculosus*. We sampled invertebrate communities from one site during 3 years of ongoing invasion by mud crabs. During this period, the abundance of mud crabs increased from very rare to common thus providing an exceptional natural setting to detect community consequences of an invasion in its early stages. In this area, the abundance of littoral producers, such as bladderwrack stands and other associated algae, is generally controlled by herbivory (Korpinen et al.

2007b; Jormalainen and Ramsay 2009; Haavisto and Jormalainen 2014). In particular, herbivory may counteract the effects of eutrophication by suppressing blooms of fast-growing opportunistic algae (Worm et al. 1999; Lotze et al. 2000, 2001; Korpinen et al. 2007b). Furthermore, cascading trophic effects of predatory fish on the abundance of producers have been documented in the Baltic Sea (Korpinen et al. 2007a; Eriksson et al. 2011; Sieben et al. 2011a, b), suggesting that upper trophic levels have the capability to regulate the level of herbivory. We thus hypothesized that mud crabs may affect the community composition of benthic invertebrates, and as a result decrease the abundance of herbivorous taxa with consequent effects on producer communities. When this occurs, the mud crab invasion may then change the functioning of the rocky littoral ecosystem.

Methods

Study area and sampling setup

We conducted the study on a rocky littoral shore in the inner Archipelago Sea of coastal Finland (60°14.6′N, 21°59.2′E). The site has a dense stand of *F. vesiculosus* (hereafter *Fucus*), extending from ~0.5 to 2–2.5 m depth. The stand has been present at least since the mid-1980s and has since been visited regularly to sample *Idotea balthica* for research purposes (by V. Jormalainen). The site is characterized by a rocky shore community assemblage typical of the area, with the foundation species *Fucus* with associated filamentous and epiphytic algae providing sheltered habitat for invertebrates and fish.

We set up an underwater garden of *Fucus* in early May 2012 for the purpose of monitoring growth and studying certain community genetics aspects of the species and the associated community. Herein, we used the garden to sample *Fucus*-associated invertebrate communities. To create the garden, we collected 30 large *Fucus* individuals, split them into five equal-sized pieces (mean ± SD wet weight: 4.56 ± 1.82 g), and attached the pieces to concrete anchors that were placed in a rocky littoral at ~1–1.5 m depth. The anchors were arranged in five plots, so that each plot contained one piece from each *Fucus* individual, totaling to 150 algal pieces. Within a plot, the algae were separated by a distance of 0.5–1 m, while the adjacent plots were separated by about 10–20 m. The study site harbored a natural *Fucus* stand, and therefore all our attached algae were within the natural stand, and thus quickly colonized by the natural fauna.

We sampled the invertebrate community associated with each algal piece repeatedly every fall: 10th–13th Sept. 2012, 18th–25th Sept. 2013 and 22nd–23rd Sept. 2014, by SCUBA-diving. In order to prevent the escape of invertebrates a diver enclosed each algal piece individually in a 1-mm mesh-net pouch, which then was received on shore and individually kept in a small plastic container filled with seawater. Back in the laboratory, we weighed the algae, and counted and identified their associated invertebrates (sedentary barnacles and bryozoans were excluded). Identification was done to the species level or lowest possible taxon. The algae were then returned to the garden for the following sampling.

To monitor the growth of *Fucus*, we counted the change in the number of vegetative apical meristems each year. The first measurement of each year was conducted in 7th–8th May 2012, 7th–9th May 2013, and 29th April–7th May 2014, and the second one while sampling the invertebrate communities in September. *Fucus* grows by elongation and branching of these apical meristems. While a part of these transform to reproductive receptacles that grow, mature and die, with self-defoliation of the thallus carrying the receptacle, the numbers of vegetative apices determine the future growth and reproductive potential of the alga. We therefore only considered the vegetative apical meristems when estimating the growth rate. In 2014, epiphytic filamentous algae on *Fucus* were very abundant, and thus we estimated their percent coverage on the *Fucus* thalli using 10 % intervals. In May, the epiphytic algae were mainly *Pilayella littoralis* which has an early summer bloom in the area, while in September, the dominant epiphytic species were *Ectocarpus siliculosus*, *Cladophora* sp. and *Ceramium tenuicorne*.

Analysis of invertebrate communities

We analyzed the differences in the bladderwrack-associated invertebrate communities among years with a permutational MANOVA, using the Bray-Curtis similarity as a measure of distance (Primer 6 with PERMANOVA + 1.0.6. software). Year was treated as a fixed factor, and the algal individual, the plot and the year-by-individual and the year-by-plot - interactions as random factors. We conducted the analysis for the raw abundance data, for the standardized data (to control for the differences in the size of the algae), and for the 4th-root transformed data (to down-weight the influence of the most dominant taxa) with no qualitative differences in the strong effect of the year on the community structure. We therefore present only the analysis of the raw data. We visualized the differences among the years with a non-metric multi-dimensional scaling (MDS) ordination plot.

To evaluate the differences in species richness among years, we calculated the Chao 1 estimate (Gotelli and Colwell 2011) that estimates the asymptotic species richness and is based on the numbers of singleton and doubleton species in the samples.

As the multivariate analysis revealed yearly differences in the invertebrate community structure, we calculated the contribution of each taxon to the differences between the years with low (2012, 2013) and high (2014) abundance of *R. harrisii*, using the SIMPER analysis in Primer 6. We further analyzed the differences among years in the abundance of *R. harrisii* and each taxon that contributed to more than 10 % of the differences between years. For this we used a generalized linear model, with errors following either a Poisson (*R. harrisii*) or a negative binomial (all other taxa) distribution, implemented by the procedure GLIMMIX in SAS 9.2 statistical software (SAS Institute Inc. 2008). Differences in the size of the sampling unit were taken into account by using the weight of the alga as the weighing variable in the analysis. We used year as a fixed factor and incorporated the sampling design by including the bladderwrack individual and the plot as random factors. After the significant effect of year, we further tested the differences between each pair of years using Tukey-adjusted pairwise tests.

For the communities in 2014 (when *R. harrisii* was abundant), we tested whether the occurrence of the crab in the sampled *Fucus* (present or not) affected the associated community using a PERMANOVA as described above, with the occurrence of crab as a fixed factor and the algal individual and the plot as random factors. With the down-weighing of the most dominant taxa (square- or 4th-root transformed data), the analysis indicated differences in communities between the algae inhabited and uninhabited by crabs, which were further analyzed using generalized linear models performed separately for each taxon (as described above).

Analysis of *Fucus* growth rate

We estimated growth rate as the change in the numbers of vegetative apices during the most intense growth season (May to September, total of 19–20 weeks), divided by the initial number of vegetative apices in May. We analyzed growth rate differences among years using a mixed model ANOVA with repeated measures (yearly growth rates as repeated measures) and plot and bladderwrack individual as random factors. Following a significant effect of year, we further tested the differences between each pair of years using Tukey-adjusted pairwise tests.

Results

The mud crab *R. harrisii* invaded the community during the 2012–2014 period. While in 2012 only two single mud crabs were found in 101 sampled algae,

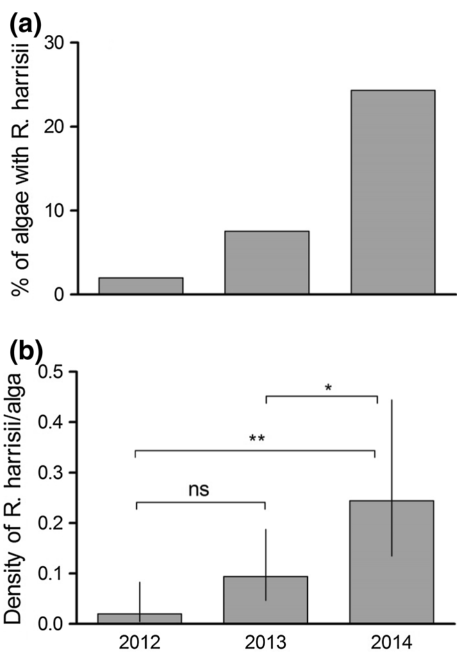

Fig. 1 The proportion of *F. vesiculosus* thalli harboring one or more *R. harrisii* individuals (**a**), and the density (mean with 95 % confidence limits) of *R. harrisii* (**b**) in 2012–2014

their total numbers increased so that in 2014 a fourth of all the algae were inhabited by one or more mud crabs (Fig. 1a). One single alga was typically inhabited by one or sometimes two mud crabs (Fig. 1). The mud crab density was statistically significantly higher in 2014 than in the two preceding years (Fig. 1b).

The structure of the invertebrate community differed among years (Table 1), and all years differed significantly from each other (2012 vs. 2013: t = 3.44, $P < 0.001$; 2012 vs. 2014: t = 5.70, $P < 0.001$, 2013 vs. 2013: t = 4.30, $P < 0.001$). Although the ordination of the communities indicated that there was some difference between the years 2012 and 2013, the main shift however occurred in 2014 when mud crabs were most numerous (Fig. 2). Communities in the years 2012 and 2013 were more similar to each other (average similarity 40.3 %) than to the community in 2014 (average similarity; 2012 vs. 2014; 15.4 %; 2013 vs. 2014: 23.7 %).

In 2014, the species richness was lower than in the two previous years (Fig. 3a): the isopod *Jaera* spp., the opistobranch *Limapontia capitata*, and the chironomid larvae were missing completely. The latter were particularly abundant in 2013 (mean 6.08 individuals/alga, 95 % confidence limits: 2.82–13.1). Together with the drop in species richness, the evenness slightly increased (Fig. 3b) while the diversity decreased (Fig. 3c). In 2014 the Shannon-Wiener *H'* was only about half of that in 2012–2013 (Fig. 3c).

The increasing abundance of mud crab itself contributed somewhat to the change in community structure. However, overall mud crab abundance remained low as compared to the dominant taxa, hence their influence on community structure was marginal. In fact, the results of permutation MANOVA (Table 1) as well as MDS-ordination (Fig. 2) remained nearly identical when conducted on data where mud crabs were removed. Thus, the shift in the invertebrate community structure can be mainly explained by the changes in abundance of dominant taxa and not by the abundance of mud crabs themselves. The most pronounced changes in the low and high mud crab abundance between years occurred due to the abundance of the gastropods *T. fluviatilis* and *Hydrobia* spp., as well as the amphipods and isopods (*Idotea* spp.) (Table 2).

Table 1 Permutational MANOVA statistics for the differences in bladderwrack-associated invertebrate communities among the years 2012, 2013 and 2014

Source	*df*	MS	Pseudo-F	*P*
Year (fixed)	2	54,578	18.8	<0.001
Alga individual (random)	29	1966	1.56	<0.001
Plot (random)	4	3245	2.57	<0.001
Y × A (random)	52	1689	1.34	<0.01
Y × P (random)	8	2699	2.14	<0.001
Residual	193	1260		
Total	288			

The four above-mentioned taxa had their lowest abundances in 2014. The gastropods and isopods steadily decreased in abundance (Fig. 4a, b, f). Overall, idoteids were treated as a genus, because not all individuals were identified to species level every year. However, *I. balthica* was the dominant species in 2012 and 2013 when they comprised at least 51 and 77 %, respectively, of the idoteids. The abundance of amphipods (including *Gammarus* and *Leptocheirus* species) increased in 2013 but ultimately decreased to their lowest abundance in 2014 (Fig. 4e). The mussels, *M. trossulus* and cardiids *Cerastoderma*/*Parvicardium* showed a different pattern: their abundance increased in 2013 but returned to the relatively low level of 2012 in 2014 (Fig. 4c, d).

In 2014 when mud crabs inhabited about 25 % of the algae, the communities differed between the algae occupied and unoccupied by mud crabs (PERMANOVA, square root transformed data: pseudo-$F_{1,\ 38} = 3.4$, $P = 0.02$), which was provoked by differences in gastropods and cardiids. Both taxa were more present in algae that were occupied by a mud crab (*T. fluviatilis* gastropods [mean, (95 % confidence limits): 0.59 (0.22, 1.6) vs. 0.09 (0.04, 0.22)] and *Cerastoderma*/*Parvicardium* cardiids [1.43 (0.483, 11.6) vs. 0.32 (0.108, 2.76)]). The densities of other taxa did not differ between algae with and without mud crabs.

The growth rate of *Fucus*, in 2012–2013, remained on a level of about five new apical meristems produced per each initial meristem, but in 2014 when herbivores were few in number, the growth rate decreased about 60 % (Fig. 5). At the same time, filamentous epiphytic algae were highly abundant on *Fucus* thalli, both in May 2014 (mean ± SD coverage: 46 ± 29 %, n = 88; with 10 % of the algae exceeding 90 % coverage) and in September 2014 (mean ± SD coverage: 22 ± 22 %, n = 74; with 10 % of the algae exceeding 50 % coverage).

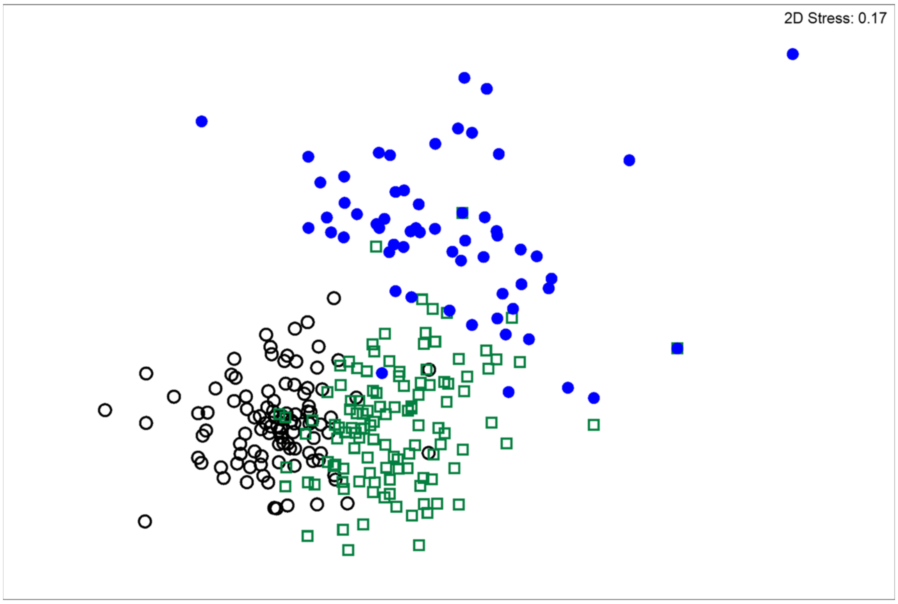

Fig. 2 MDS ordination of the bladderwrack-associated invertebrate communities in 2012 (*open circle*), 2013 (*open square*) and 2014 (*closed circle*)

Discussion

With the invasion of the mud crab the *Fucus*-associated invertebrate community underwent a major quantitative and qualitative transition: the diversity dropped in the third year of invasion as a consequence of the decreasing abundance and loss of certain taxa: From 2012 to 2014 the total density of gastropods decreased 99 % and that of crustaceans 75 %. The former community dominated by gastropods and crustaceans shifted to a mussel dominated community with an overall low abundance of invertebrates. Even though invasive crabs can profoundly influence the abundance of native invertebrates (e.g. decreases in amphipod abundance) (Grosholz et al. 2000), this is the first time that such effects have been reported for the invasive *R. harrisii* in field conditions with natural availability of prey items.

We suggest that the shift in invertebrate community structure was caused by the mud crab invasion for the following reasons. First, *T. fluviatilis* and *Hydrobia* spp. gastropods, amphipods and isopods, as well as chironomids, are the typical dominant invertebrate groups in littoral macrophyte stands of the Archipelago Sea and elsewhere in the Northern Baltic Sea (e.g. Boström et al. 2006; Wikström and Kautsky 2007; Korpinen et al. 2007b; Haavisto and Jormalainen 2014; Gagnon et al. 2015). This is also true for our experimental site, which has been visited regularly since the early-80s to collect isopods for research purposes. This non-quantitative sampling was only focused on idoteids and therefore no time series data on community composition exist. However, the very low abundances of these taxa, which coincided with the increasing abundance of mud crabs, are unusual. Second, laboratory feeding experiments have verified that *R. harrisii* feeds on gammarid amphipods, *Idotea* spp. as well as *T. fluviatilis* gastropods and *M. trossulus* bivalves, while in a caging experiment conducted in the field, *R. harrisii* also decreased the number of *T. fluviatilis* on bladderwrack (Forsström et al. 2015). Remains of amphipods have also been found in the stomachs of field-caught mud crabs (Hegele-Drywa and Normant 2009). Third, other crab species with a functional role similar to *R. harrisii* are lacking from the area, and, thus, anti-predatory adaptations such as avoidance behaviors of crab predators might be missing or poorly developed in the local invertebrate fauna (Sih et al. 2010). Prey naivety, i.e. lack of co-evolved anti-predatory behaviors, has been suggested to be the principal reason explaining the strong effect of invasive consumers on their prey populations (Paolucci et al. 2013). Thus, the shift in the invertebrate community structure into a very distinctive one as typically found in the area coincided with the invasion of a species that is a novel kind of a predator and capable of preying on the dominant taxa. We consider the mud crab invasion as

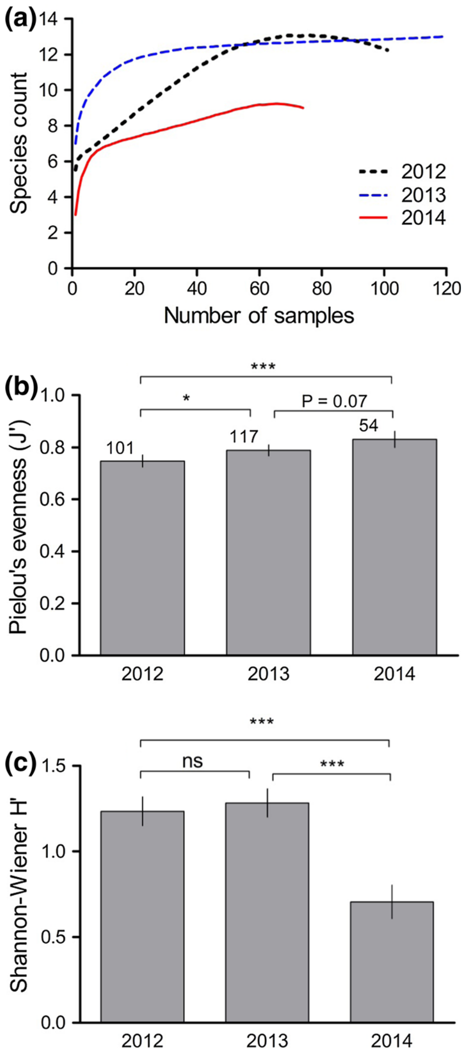

Fig. 3 Chao 1 estimates of the number of species (**a**), Pielou's evenness (**b**), and Shannon–Wiener diversity (**c**) in the bladderwrack-associated invertebrate communities in 2012–2013

the most likely explanation of the changes in the invertebrate community, but as always with observational data, we cannot exclude the possible influence of other biotic or abiotic factors.

The lower abundance of gastropods, idoteids, and amphipods following the establishment of the mud crab population could be due to direct predation and numerical effects, or, to non-consumptive effects (i.e. prey avoiding the predator by fleeing from algae which are inhabited by crabs). Even non-consumptive effects of predation have been shown to be important as predator-induced shifts in habitat choices may modify consumer abundance (Vonesh et al. 2009) and producer biomass (Reynolds and Sotka 2011). As invertebrates did not avoid algae inhabited by mud crabs, and all the algae, including those without any crabs, had much lower prey abundances in the year with high mud crab abundance than in the years with low mud crab abundance, we suggest that direct predation is the more likely explanation here. The gastropod *T. fluviatilis* and the cardids *Cerastoderma/Parvicardium* were even more abundant in algae occupied by mud crabs implying that the highly mobile mud crabs actively move into algae inhabited by these species. A direct predation effect is also supported by laboratory experiments, showing that the mobile crustaceans *I. balthica* and gammarid amphipods are readily predated by mud crabs and are preferred over blue mussels (Forsström et al. 2015). Furthermore, *I. balthica* is observed to lack the antipredation behaviors it typically uses against fish when faced with this novel predator (unpublished data by Maria Yli-Renko, pers. communication).

The shift in the invertebrate community may have far-reaching consequences on the ecosystem function of rocky littoral macroalgal assemblages. These arise

Table 2 Average abundance of the most common taxa and their contribution to the differences in community composition between the years with low (2012–2013) and high (2014) abundance of *R. harrisii*

Taxon	Average abundance (# alga^{-1})			Contrib. to differences (%)	
	2012	2013	2014	2012–2014	2013–2014
T. fluviatilis	22.5	7.2	0.4	47.7	21.7
Hydrobia spp.	7.9	0.3	0.0	12.5	0.9
Amphipoda	6.9	12.9	1.4	13.7	37.6
Idotea spp.	4.4	0.9	0.04	10.4	3.0
M. trossulus	4.1	4.8	4.1	10.0	17.0
Cerastoderma/Parvicardium	1.6	3.0	1.0	4.3	10.1
Jaera spp.	0.04	1.8	0.0	0.1	5.5

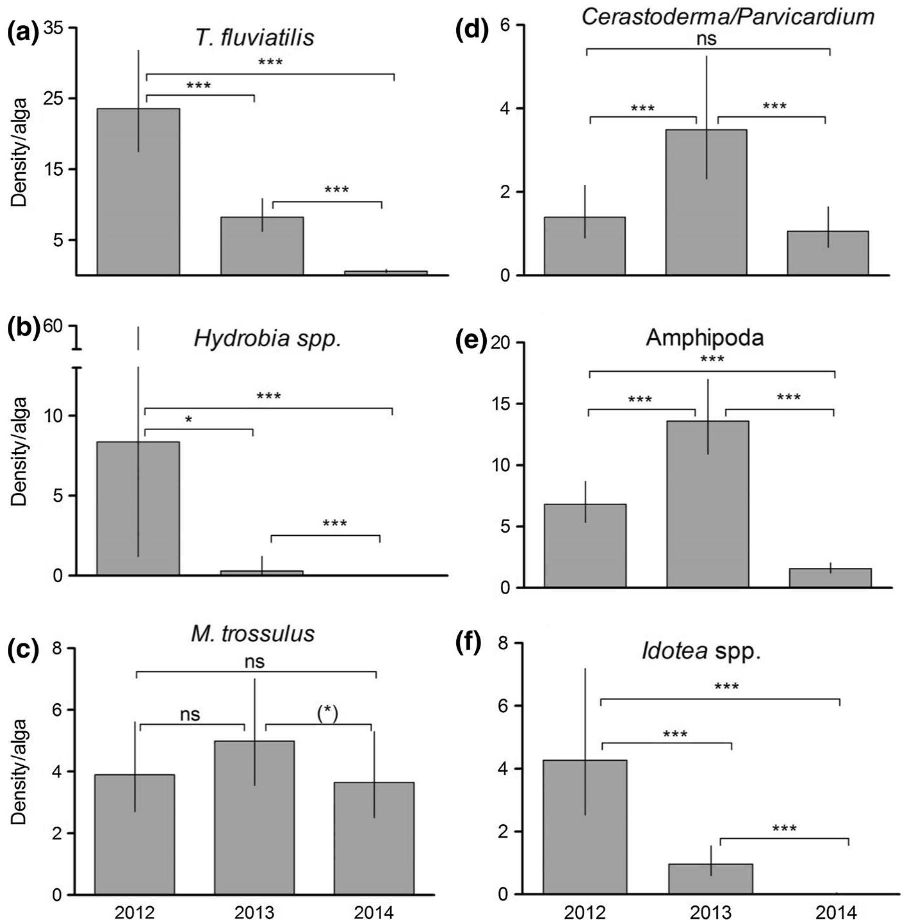

Fig. 4 Densities (mean with 95 % confidence intervals) of four mollusk taxa (**a**–**d**) and two crustacean taxa (**e**, **f**) during the years with low (2012–2013) and high (2014) abundance of *R. harrisii*. Pairwise Tukey-adjusted differences between years are indicated as follows: ns $P > 0.01$; *$P < 0.05$; **$P < 0.01$; ***$P < 0.001$

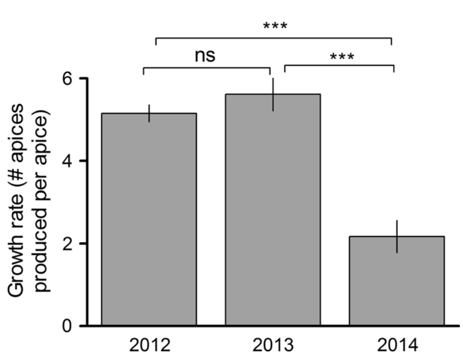

Fig. 5 Growth rate of *Fucus*, in terms of new apical meristems produced per initial apical meristem (mean with 95 % CIs), during May–September in the years with low (2012–2013) and high (2014) abundance of *R. harrisii*. Pairwise Tukey-adjusted differences between years are indicated as in the Fig. 4

through changes in the strength of producer–herbivore interaction, caused by mud crab predation on the dominating grazer taxa. This interaction is a major determinant of ecological function in ecosystems, i.e. productivity and energy flow from producers to higher trophic levels. This is particularly true in rocky littoral habitats, where the amount of the biomass removed by herbivory is the highest among all marine littoral habitats globally (Poore et al. 2012). Therefore, herbivory can be expected to have a major influence on the abundance and structure of producer communities.

Herbivory is severe in *Fucus*-dominated Baltic Sea rocky littoral habitats as all producer groups—microalgae, filamentous macroalgae and *Fucus* itself—are grazed by some taxa of the highly abundant herbivores. Gammarid amphipods are particularly effective in controlling filamentous macroalgae (Goecker and Kall 2003; Eriksson et al. 2011), while herbivory on early recruitment stages of macroalgae is mainly due to gastropods (Malm et al. 1999; Korpinen et al. 2007a) and gammarid amphipods (Eriksson et al. 2011), both of which also feed on periphytic microalgae and colony-forming diatoms. Herbivory in the early recruitment stages of macroalgae is intense, for example, nearly 80 % of the early recruits of *Fucus* can be removed by herbivores (Korpinen et al. 2007b). Herbivory on adult thalli of *Fucus* in the Northern Baltic Sea is almost exclusively by the isopod *Idotea balthica*: in autumn, when its densities can reach dozens to well over 100 individuals per average sized bladderwrack thallus (Korpinen et al. 2010) it is capable of removing 60–70 % of the bladderwrack biomass, while the intensity of herbivory decreases with lower densities during the winter (Jormalainen and Ramsay 2009; Haavisto and Jormalainen 2014). Occasional complete defoliations of bladderwrack stands by this herbivore have been observed in the Baltic Sea (Kangas et al. 1982; Haahtela 1984; Rönnberg et al. 1985; Engkvist et al. 2000; Nilsson et al. 2004).

Grazing is also important because it can affect community composition of producers thereby counteracting the effects of eutrophication. For example, Worm et al. (1999) found that experimental nutrient enrichment leads to decreased diversity of macroalgae, but only when herbivores were excluded. Similarly, Korpinen et al. (2007b) found that herbivores efficiently counteracted the increase of opportunistic filamentous algae that otherwise flourished in nutrient-enriched conditions. Furthermore, gastropods that feed on periphyton can have positive effects on perennial macroalgae when they feed on the thallus and remove epibiota that compete for light and nutrients with the basibiont (Jormalainen et al. 2003). For example, the removal of epibiotic periphyton by *T. fluviatilis* from *Fucus* thalli resulted in a doubled growth rate compared to counterparts without snails (Honkanen and Jormalainen 2005). Hence, grazing, although occurring on opportunistic as well as perennial species, benefits the latter by diminishing competition and preventing their competitive exclusion.

We found that, in 2014, the growth rate of *Fucus* dropped to less than half of what it used to be. At the same time, mud crabs were abundant while the abundance of herbivorous invertebrate taxa was very low, and filamentous opportunistic algae prospered on the *Fucus* thalli. This suggests that *Fucus* suffered from the competition by filamentous algae, which were able to increase because of lack of herbivore control.

Although mud crabs are omnivores and plant material and detritus is often found as part of their diet (Hegele-Drywa and Normant 2009), our findings suggest that their major ecological role is predatory and their predatory effects may cascade to the level of producers. Owing to the strong regulatory role of herbivory in the littoral of the Baltic Sea, the influence of the mud crab on the littoral ecosystems may far outweigh the numerical effect it has on invertebrate communities. Our data gives support for the following hypothesis, although direct experimental evidence is still needed: by forcing the herbivore- and deposit-feeder-dominated invertebrate community into a mussel-dominated filter-feeder community, mud crabs may fundamentally change ecosystem function. As a consequence of the weakened or missing producer–herbivore link, producer biomass will accumulate. This will favor fast-growing filamentous algae thereby increasing the amount of decaying macroalgal biomass and drifting algal mats, the decomposition of which will augment benthic hypoxia and anoxia (Vahteri et al. 2000; Arroyo et al. 2012). Biomass accumulation of filamentous algae occurs at the cost of the perennial, habitat-forming foundation species *F. vesiculosus*, which leads to less diverse littoral habitats with reduced ecosystem services provided by the perennial aquatic macrophyte stands (e.g. sequestration of carbon and nutrients, provision of nursery grounds for fish reproduction).

The ecological consequences of rapidly intensifying coastal invasions can be profound (Grosholz 2002). However, so far indirect biotic effects such as the modification of trophic cascades and their ecosystem consequences have been overlooked in invasion biology (White et al. 2006). Our awareness about the role of animals at high trophic levels in ecosystem functioning is strengthening (Estes et al. 2011; Schmitz et al. 2014), and the most influential effects may often be the indirect ones arising through the trophic cascades. Thus, in order to predict the ecological consequences of invasions, we need to have good knowledge on the specific influence of invasives on indigenous communities as well as an understanding of the biotic interactions and functional roles of the species involved.

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